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Return of Topeka Shiner (*Notropis topeka*) to Restored Oxbows in the White Fox Creek Watershed, Iowa, USA

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The Topeka shiner (*Notropis topeka)*, a federally endangered species of minnow endemic to the Great Plains region of the central United States, has experienced widespread population declines resulting from loss of habitat. *N. topeka* habitat in Iowa, most notably oxbow wetlands, was mostly eliminated from the landscape during European settlement and agriculturalization of the region. Over the last two decades, restoration of oxbow habitats in Iowa has been increasing. Restorations provide critical habitat for *N. topeka* and regenerate a variety of ecosystem services that benefit many species of flora and fauna. There are signs that restoration of oxbow ecosystems is generating positive impacts on *N. topeka* recovery. Recent studies revealed that *N. topeka* populations are recovering within a few Iowa watersheds. In the investigation described here, we report the 2020 discovery of two *N. topeka* specimens in different restored oxbows within the White Fox Creek HUC10 of north central Iowa. Prior to these collections, the species had been undetected within this basin for 36 years. (It is possible that *N. topeka* persisted within the basin but remained undetected during sampling efforts.) Multiple oxbows have been restored within the White Fox Creek basin in recent years, and significant source populations of *N. topeka* can be found in nearby watersheds. These collections suggest that as oxbow restorations continue and land stewardship practices improve, *N. topeka* populations might recover and become reestablished within the historical range.

KEYWORDS: Topeka shiner, *N. topeka*, restoration, oxbow, endangered, conservation

Introduction

The Topeka shiner (*Notropis topeka)*, a federally endangered species of minnow (family Leuciscidae; Gidmark and Simons 2014, Tan and Armbruster 2018), is endemic to the Great Plains region of the central United States (Tabor 1998). Historically, the geographic range of this species included Iowa, Kansas, Minnesota, Missouri, Nebraska, and South Dakota, where it inhabited low-order and headwater prairie streams (Tabor 1998, Wall and Berry 2004, Mammoliti 2014). Areal cover of tallgrass prairie, originally the dominant land cover in the Great Plains, has declined by 82 to 99% since European settlement of the region (Samson and Knopf 1994). This habitat loss has resulted in dramatic population declines in both terrestrial and aquatic plant and animal species, including *N. topeka* (Samson and Knopf 1994, Murkin 1998, Dodds et al. 2004, Gido et al. 2010). Currently, *N. topeka* occupies approximately 20% of its historical range (Tabor 1998, Wall and Berry 2004, Wall et al. 2004). Within Iowa, the range of *N. topeka* has contracted by 40%, and the species may have been extirpated from 3 of the 20 hydrologic unit code 10 (HUC10) watersheds where it historically occurred (Pierce et al. 2019).

Recent studies determined that oxbow wetlands are critical habitat for *N. topeka* (Bakevich et al. 2013, Bakevich et al. 2015, Simpson et al. 2019). Oxbow wetlands are floodplain ecosystems that form when a river meander separates from the main channel through natural erosional processes. Throughout most of the geographic range of *N. topeka*, streams were channelized and straightened during the early 20th century, eliminating many of these off-channel habitats (Dodds et al. 2004, Costigan and Daniels 2012). It is likely that the loss of oxbow wetlands has contributed to the decline of *N. topeka*.

Over the last two decades, increased emphasis has been placed on restoring oxbows to benefit *N. topeka* recovery (Fig. 1). Oxbows are restored by excavating topsoil down to the former stream bed depth, which results in a reestablished groundwater connection, leading to greater water depths and volume (Kenney 2013, Simpson et al. 2019). In addition, some recent oxbow restorations have included the construction of a narrow connection to the adjacent stream on the downstream end of the oxbow to allow for connection to the stream during 1.5 to 2-yr flood events (Schilling et al. 2019). The connection is vital, as it allows fishes to move between the oxbow and associated stream during flood events that normally would not have inundated the oxbow. Oxbows also provide habitat for a variety of flora and fauna, sequester nutrients such as nitrogen and phosphorus, create recreation opportunities such as angling, provide floodwater retention, and increase the aesthetic value of the floodplain (Jones et al. 2015, Schilling et al. 2019). The cost of these restorations is similar to other edge-of-field practices to improve water quality, such as woodchip bioreactors; but the ecosystem services provided by oxbows surpass those of many these practices (Shilling et al. 2019).

There are signs that restoration of oxbows is having a positive impact on *N. topeka* populations. Pierce et al. (2019) determined that distribution of *N. topeka* in Iowa is increasing after several decades of decline and reasoned that a potential cause of this improvement is the restoration of oxbows. Simpson et al. (2019) discovered that *N. topeka* was present more often and with higher average relative abundances in restored oxbows than in unrestored oxbows. Here we present anecdotal evidence further supporting the notion that oxbow restoration is benefitting recovery of *N. topeka*: our discovery of *N. topeka* in two restored oxbows within the White Fox Creek HUC10 in central Iowa during 2020.

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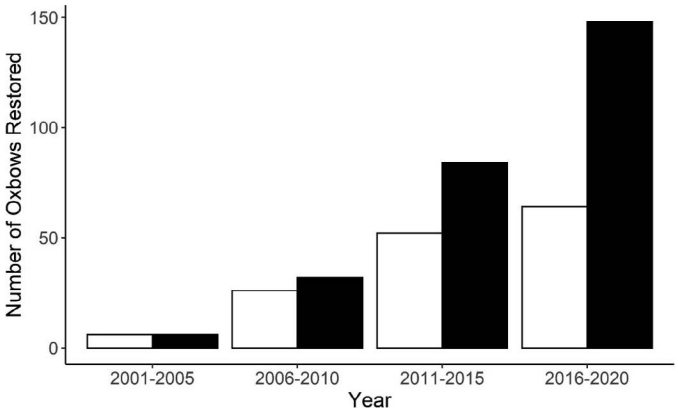


Fig. 1. Number of oxbows restored in Iowa per five-year period, 2001 to 2020. White bars represent oxbows restored during the time period, black bars represent cumulative total (A Kenney, US Fish and Wildlife Service, pers. comm.; D Weissenfluh, US Fish and Wildlife Service, pers. comm.; K Wilke, The Nature Conservancy, pers. comm.).

Methods

We conducted a survey of Topeka shiner (*N. topeka)* geographic distribution as part of a study comparing effectiveness of different sampling methods for characterizing oxbow fish communities (Osterhaus et al., in preparation). During the early summer of 2020 (mid-May through mid-June), the fish communities of 12 restored oxbows were sampled. Ten of these oxbows were located within the Boone River hydrologic unit code 8 (HUC8); five within the White Fox Creek HUC10, five within the Lower Boone River HUC10 (Fig. 2). The other two oxbows were located within the Cedar Creek–Upper North Raccoon River HUC10 of the North Raccoon River HUC8 (Fig. 3). All of the sampled oxbows are located within the Des Moines River basin (Upper Mississippi River drainage; Fig. 3). Each of the 12 oxbows was sampled four times over a four-week period (48 total site visits) by using each of the following sampling methods once at each oxbow: backpack electrofishing, fyke netting, minnow trapping, and seining. See Osterhaus et al. (in preparation) for a detailed description of fish sampling methods.

In summary, backpack electrofishing was conducted within all wadeable oxbow habitat. Fyke netting was conducted by placing three unbaited mini-fyke nets (4.0 m lead, 0.6-cm mesh, largest hoop opening = 0.6 m×1.2 m) perpendicular to the shoreline within the open water zone at points spaced evenly along the length of the oxbow. Fyke nets were deployed and retrieved after 24 h. Minnow trapping involved deployment of four un-baited minnow traps (2.5 cm opening, 6.35 mm mesh, 40 cm long × 22 cm diameter) for 24 h at points evenly spaced along the length of the oxbow. Seining was conducted by a single pass of a bag seine (10.7 m×1.8 m, 6.35 mm mesh or 16.8 m×1.8 m, 6.35 mm mesh depending on oxbow width), covering the length of the oxbow. For oxbows less than 16 m in average width, the 10.7 m wide seine was used. For oxbows greater than 16 m in average width, the 16.8 m seine was used. Regardless of sampling method, all fish were handled with care to minimize stress as they were identified, measured for length, counted, and released alive.

Results

In total, 32,862 fish were collected from the 12 oxbows sampled, representing 26 species (Osterhaus et al. in preparation). Two Topeka shiner (*N. topeka)* were collected from the 12 oxbows sampled–one each from the two restored oxbows within the White Fox Creek HUC10 (Table 1). At oxbow one (White Fox Creek HUC10, 11 km northeast of Webster City, Iowa), a single *N. topeka* was collected on 29 May 2020 via fyke netting. This *N. topeka* was identified as an adult female given the lack of breeding coloration that is present on males of this species during the breeding season (mid-May to early August; Dahle 2001, Witte et al. 2009). At oxbow two (White Fox Creek HUC10, 8 km northeast of Webster City, Iowa), a *N. topeka* was collected on 10 June 2020 via seining. The *N. topeka* was identified as an adult male given the presence of breeding coloration and tubercles covering the head and nape.

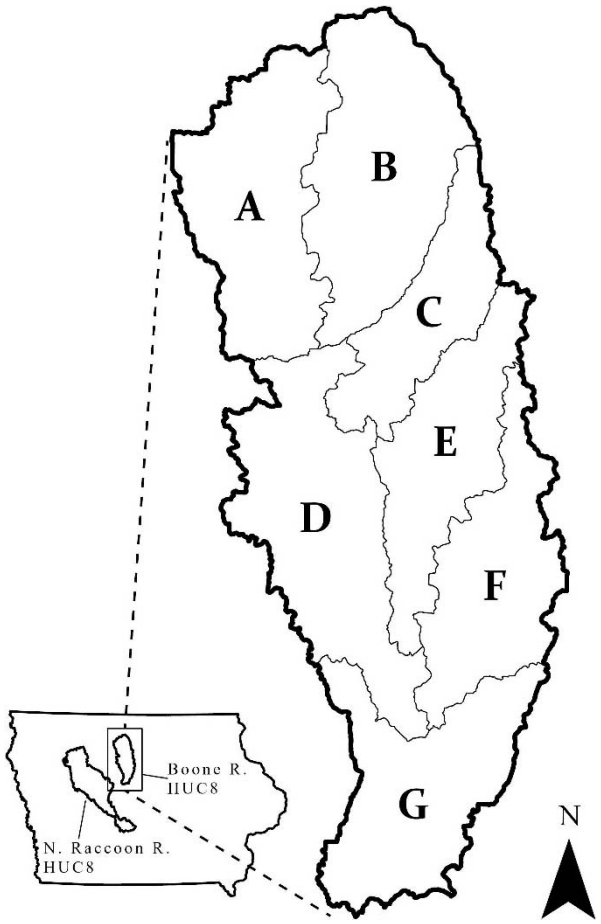


Fig. 2. HUC10 watersheds within the Boone River HUC8, Iowa, USA. (A) Prairie Creek; (B) Upper Boone River; (C) Otter Creek; (D) Middle Boone River; (E) Eagle Creek; (F) White Fox Creek; (G) Lower Boone River.

Oxbow one, restored in 2012, is a typical prairie oxbow with low bank angles, lack of overhead canopy cover, and an abundance of emergent and submerged aquatic macrophytes (> 50% total cover). Measuring 138.7 m in length and an average width of 15.3 m, oxbow one is the largest of the 12 oxbows sampled during the study (Table 1). During high flow events, oxbow one connects with two adjacent oxbows and the mainstem of White Fox Creek. Water levels were high at oxbow one throughout the sampling period, with a small connection existing between the oxbow and the stream during two sampling events.

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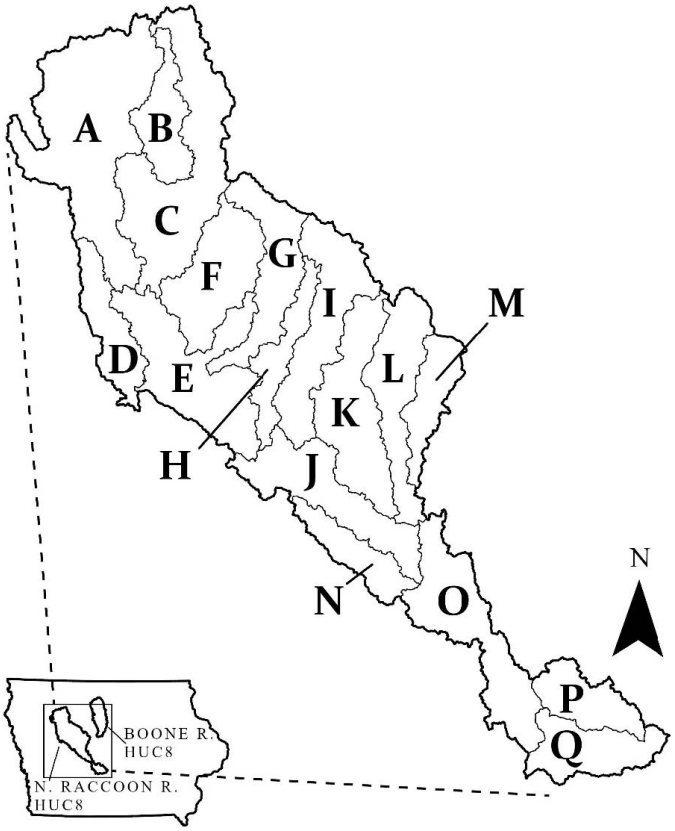


Fig. 3. HUC10 watersheds within the North Raccoon HUC8, Iowa, USA. (A) North Raccoon River (B) Little Cedar Creek (C) Cedar Creek – Upper North Raccoon River (D) Indian Creek (E) Upper North Raccoon River (F) Camp Creek (G) Lake Creek (H) Purgatory Creek (I) Cedar Creek – Middle North Raccoon River (J) Middle North Raccoon River (K) Hardin Creek (L) Buttrick Creek (M) East Buttrick Creek (N) Greenbrier Creek (O) Lower North Raccoon River (P) Walnut Creek (Q) Raccoon River

Oxbow two, restored in 2011, differs broadly from oxbow one in habitat present. Oxbow two is located within the edge of wooded riparian floodplain adjacent to the White Fox Creek mainstem. An access road for an agriculture field borders the bank along one side of oxbow two. Aquatic vegetation within oxbow two was minimal (< 5% total cover). However, as with oxbow one, oxbow two also connects to the mainstem of White Fox Creek during high flow events.

Mean fish species richness values at oxbows one and two were 8.25 and 9.00, respectively, over the four sampling sessions conducted at each oxbow, representing the two highest recorded species richness values among oxbows sampled by Osterhaus et al. (in preparation). While species richness was high at the two oxbows containing *N. topeka*, CPUE (catch per unit effort) values at oxbows one and two were below average compared to the other oxbows surveyed. Both oxbows in which *N. topeka* were collected were at the larger end (in terms of average length and width) of the range of oxbows sampled during the study (Table 1).

Discussion

Before our discovery, Topeka shiner (*N. topeka)* had not been detected in the White Fox Creek HUC10 for 36 years, despite several fish surveys conducted by Iowa Department of Natural Resources biologists (Iowa Department of Natural Resources, 2020) and Iowa State University faculty and graduate students (BW Menzel, SJ Clark, NT Simpson). Before 2020, the most recent collection of *N. topeka* within the White Fox Creek HUC10 was made by Curt Meyer and Dan Weidemeier in an unnamed tributary to White Fox Creek (6 km south of Clarion, Iowa) on 15 June 1984. This collection was made via backpack electrofishing over a stream channel comprised of sand and gravel substrate (Menzel 1984).

Given the 36-year gap during which *N. topeka* was undetected in the White Fox Creek HUC10 and the periodic sampling that occurred during this period, it is possible that the species was extirpated from the basin. It is also possible that *N. topeka* persisted within the basin but remained undetected during sampling efforts since 1984.

If we assume that *N. topeka* was extirpated from the White Fox Creek HUC10 sometime within the last 36 years, then the *N. topeka* now within the basin may have recolonized from another nearby HUC10 within the Boone River HUC8. Pierce et al. (2019) assessed the status of *N. topeka* within each HUC10 within the Boone River HUC8 and concluded that of the seven HUC10s within the Boone River HUC8, *N. topeka* may have been extirpated within the White Fox Creek HUC10, at risk within the Lower Boone River HUC10, stable within the Eagle Creek HUC10, possibly stable within the Prairie Creek HUC10 and Middle Boone River HUC10, and possibly recovering within the Upper Boone River HUC10 and Otter Creek HUC10 (Fig. 2). Fish community surveys during 2019

TABLE 1. Physical attributes and fish community measures for oxbows sampled by Osterhaus et al. (in preparation). Mean catch per unit effort (CPUE) and mean species richness (based on four sampling sessions) were not calculated for oxbow 3, as a fish kill occurred at the oxbow during the study period.

Oxbow	HUC10	Length (m)	Mean width (m)	Mean CPUE (no. sh/100m^2)	Avg Species Richness	Topeka shiner collected?
1	White Fox Creek	138.70	15.30	23.09	8.25	Yes
2	〃 〃	132.20	11.92	53.39	9.00	Yes
3	〃 〃	40.70	4.80	–	–	No
4	〃 〃	78.60	5.86	35.88	6.75	No
5	〃 〃	113.30	13.20	4.33	4.00	No
6	Lower Boone River	23.10	5.60	75.95	3.00	No
7	〃 〃	82.00	8.80	90.74	3.25	No
8	〃 〃	47.60	5.31	346.68	1.45	No
9	〃 〃	27.20	6.04	523.32	2.25	No
10	〃 〃	36.00	7.60	771.29	3.25	No
11	Cedar Creek–Upper North Raccoon River	107.70	20.40	21.37	3.75	No
12	〃 〃	101.70	15.70	97.76	8.25	No

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and 2020 within the Boone River HUC8 confirmed that *N. topeka* was present, at times in high numbers, within the watershed (833 individuals were collected from a single restored oxbow in 2020, Otter Creek; unpublished data). Given the apparently healthy *N. topeka* populations in surrounding HUC10 watersheds, there appears to be a substantial source population from which *N. topeka* could emigrate and enter the White Fox Creek HUC10.

If we assume that *N. topeka* has persisted within the White Fox Creek HUC10 but remained undetected during recent sampling efforts, then it is likely that the species persisted in low numbers within upstream reaches of the basin. Recent sampling efforts have focused primarily on oxbows along the mainstem of White Fox Creek and the main channel of the creek itself. Therefore, the discovery of *N. topeka* within this previously sampled area of the mainstem of White Fox Creek could indicate that the populations of *N. topeka* that persisted within the upper reaches of the White Fox Creek basin could be growing and expanding their distribution.

It is noteworthy that both *N. topeka* specimens discovered within the White Fox Creek HUC10 were in restored oxbows. As the number of oxbow restorations increases within the Boone River HUC8, the amount of high-quality spawning habitat present within the basin will also increase. Within the White Fox Creek HUC10 specifically, multiple oxbows were restored as recently as 2020 near the two oxbows occupied by *N. topeka*. It is possible that even if the *N. topeka* currently within the White Fox HUC10 fail to reestablish populations, the quantity and quality of habitat present within the basin, as well as the proximity to viable source populations, will create a hospitable environment for future *N. topeka* recolonization.

Although the species may have been absent from the basin for a considerable period, oxbows were restored within the White Fox Creek HUC10 to provide suitable habitat for *N. topeka* in the event of its return. The strategy of restoring oxbows within areas where *N. topeka* has occurred historically, but not recently, has also been employed within the North Raccoon River HUC10 and the Cedar Creek–Upper North Raccoon River HUC10 within the North Raccoon HUC8 (Fig. 3, Pierce et al. 2019). Similar to the White Fox Creek HUC10, these two basins are in close proximity to stable *N. topeka* populations that could serve as source populations. A key component to the recovery of *N. topeka* in Iowa is the return of the species to its historical range (US Fish and Wildlife Service 2021). These oxbows, restored in areas of historical occurrence, provide high-quality habitat for *N. topeka* if individuals enter the basin in sufficient numbers to reestablish the species. This strategy of restoring oxbows will be integral to the recovery of *N. topeka* in Iowa.

To better document how restoration activities are aiding in the recovery of *N. topeka*, *N. topeka* populations should be surveyed further, both in areas of recent occurrence and historical occurrence. The monitoring of populations within the region will allow for more informed decision-making when determining where oxbow restorations should occur and will allow for oxbows to be restored in ways that maximize their suitability for *N. topeka*. Restoration of oxbow wetlands, along with continuing improvements in land stewardship in agricultural regions, will generate positive results in the recovery of the endangered *N. topeka*.

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